



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECI1143-02 Probability & Statistical Data Analysis
Assignment 1

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Group Members:

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Question 1

(a) Population : Students from faculty of computing

Sample : randomly selected student from faculty of computing

(b) (i) Name

(i) Discrete

(ii) Categorical

(iii) Nominal : Ali, Abu, Sara

Education level of parent

(i) Discrete

(ii) categorical

(iii) ordinal : SPM, pre-university, degree, master, phd.

Hometown

(i) Discrete

(ii) Numerical

(iii) Nominal : Penang, Johor, Melaka, Ipoh

Year of study

(i) Discrete

(ii) Numerical

(iii) ordinal : Freshman, Sophomore, Junior, Senior

CGPA

(i) Continuous

(ii) numerical

(iii) ratio : 1.50, 3.00, 3.28, 4.00

Family monthly income

(i) continuous

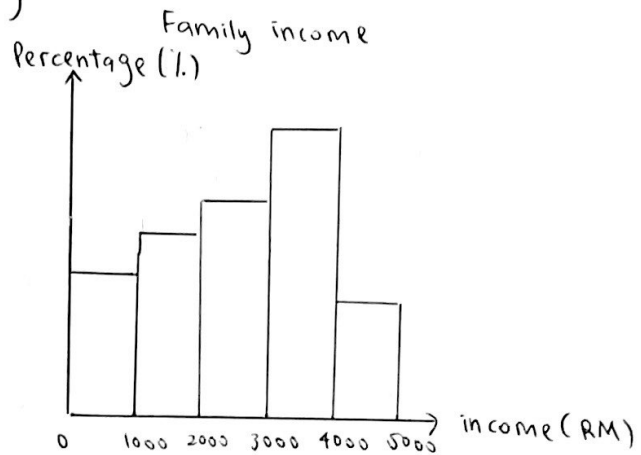
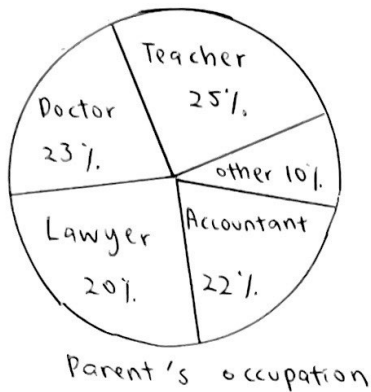
(ii) numerical

(iii) ratio : RM4700-20, RM5000, RM7300.30

(c) Questionnaire through Google Form,
random street interview at Faculty of Computing

(d) Existing database from the faculty

(e)



Question 2

Number of injured students	Frequency	relative frequency
0	3	0.1
1	2	0.07
2	7	0.23
3	7	0.23
4	5	0.17
5	4	0.13
6	2	0.07
Total	30	1.00

Q₃

① tally sheet

Weight	tabulation	frequency	weight	tabulation	frequency
450		10	462		2
451		2	463		3
452		2	464		4
453		7	465		3
454		3	466		4
455		6	467		5
456		4	468		0
457		6	469		4
458		4	470		5
459		3	471		8
460		3	472		9
461		2	473		1
			474		0

② Range

$$\begin{aligned}
 R &= X_h - X_l \\
 &= 473 - 450 \\
 &= 23
 \end{aligned}$$

③ cell interval

$$\begin{aligned}
 i &= \frac{R}{1 + 3.322 \log n} \\
 &= \frac{23}{1 + 3.322 \log 100} \\
 &= 3.00
 \end{aligned}$$

trial and error

$$\begin{aligned}
 i = 1.00; h &= \frac{R}{i} = \frac{23}{1.00} = 23 \\
 i = 3.00; h &= \frac{R}{i} = \frac{23}{3.00} = 8 \\
 i = 5.00; h &= \frac{R}{i} = \frac{23}{5.00} = 5
 \end{aligned}$$

④ cell midpoint

$$MP_k = X_k + \frac{i}{2} = 450 + \frac{3.00}{2} = 451.50$$

cell midpoint

451.50

454.50

457.50

460.50

463.50

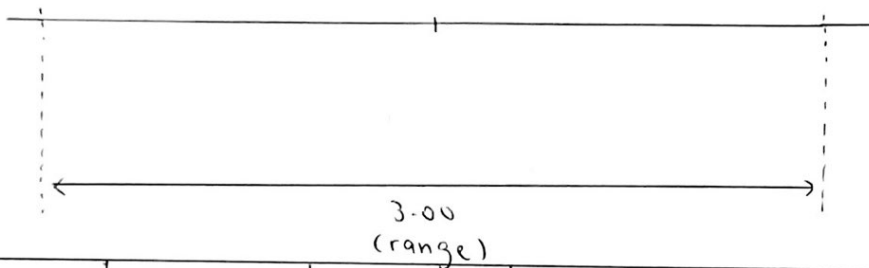
466.50

469.50

472.50

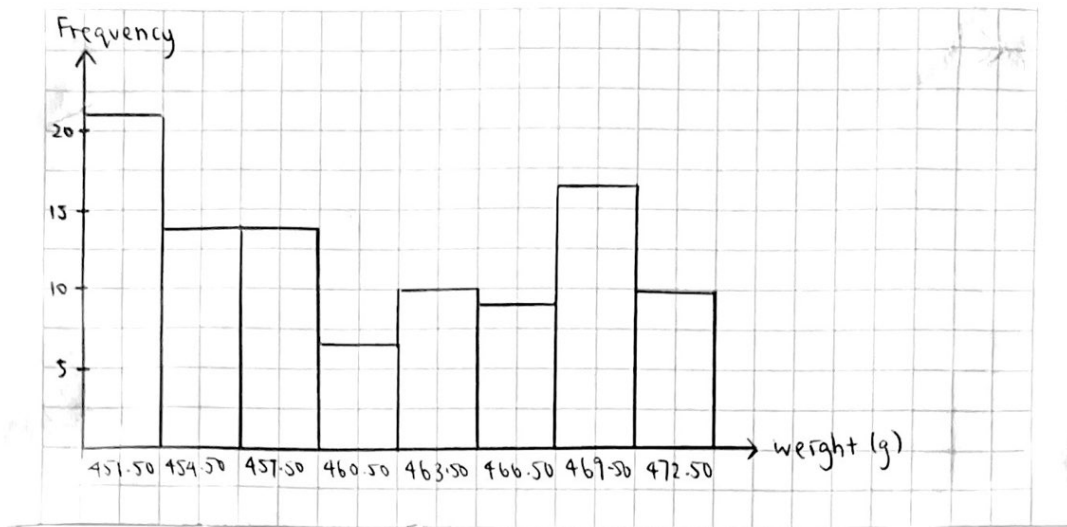
⑤ cell boundaries

lower boundary -1.5 midpoint +1.5 upper boundary
450.00 451.50 453.00

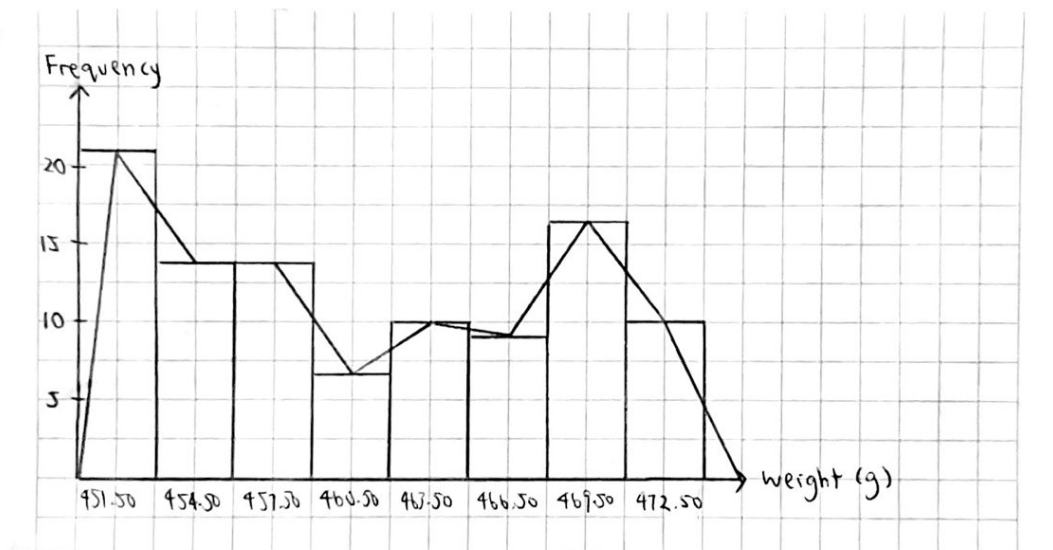


⑥ class	Boundaries	cell midpoint	Frequency	Relative frequency	cumulative frequency	cumulative relative frequency
450-453	449.5-453.5	451.50	21	0.21	21	0.21
453-456	453.5-456.5	454.50	13	0.13	34	0.34
456-459	456.5-459.5	457.50	13	0.13	47	0.47
459-462	459.5-462.5	460.50	7	0.07	54	0.54
462-465	462.5-465.5	463.50	10	0.10	64	0.64
465-468	465.5-468.5	466.50	9	0.09	73	0.73
468-471	468.5-471.5	469.50	17	0.17	90	0.90
471-474	471.5-474.5	472.50	10	0.10	100	1.00
Total			100	1.00		

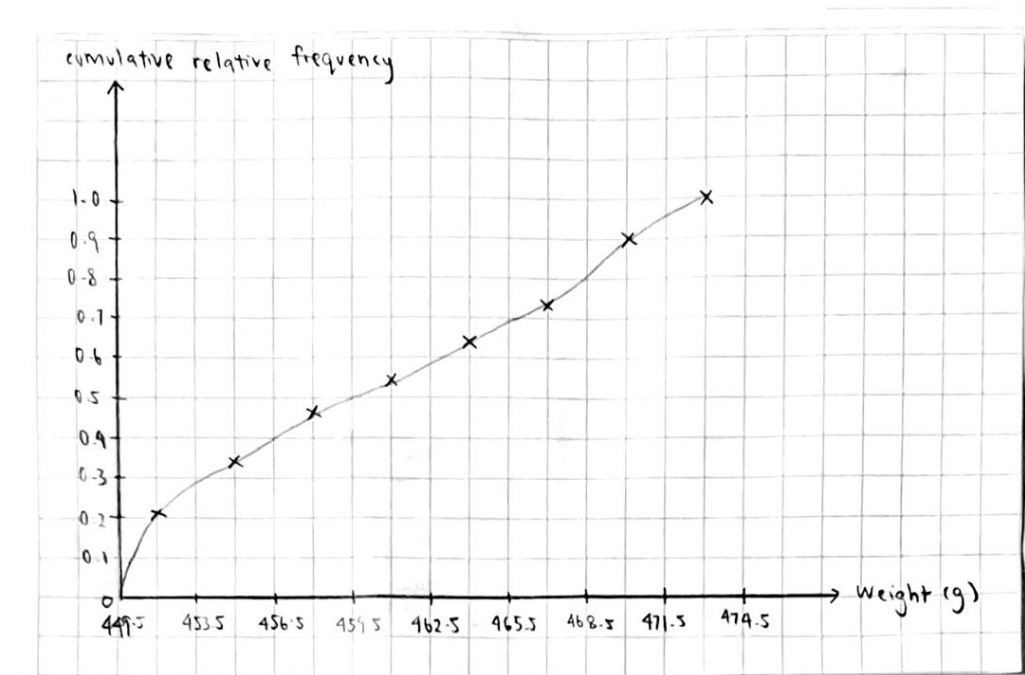
⑦ Histogram



Frequency polygon



④ Ogive



Question 4

78, 702, 765, 811, 832, 855, 896, 902, 905, 918, 919, 920, 923, 929, 936, 938, 948, 950, 956, 958, 958, 970, 972, 978, 1009, 1022, 1035, 1037, 1045, 1067, 1085, 1092, 1102, 1122, 1126, 1151, 1150, 1157, 1157, 1162, 1170, 1195, 1195, 1196, 1217, 1237, 1311, 1333, 1340

$$1^{\text{st}} \text{ quartile } (Q_1) = \frac{25 \times 50}{100} = 12.5, k = 13$$

$$\therefore Q_1 = Y[13] = 923$$

$$Q_2 = \frac{50 \times 50}{100} = 25; \quad \frac{Y[25] + Y[26]}{2} = \frac{1009 + 1009}{2} = 1009$$

$$\begin{aligned} \text{IQR} &= 1156 - 923 \\ &= 233 \end{aligned}$$

$$\therefore Q_2 = 1009$$

$$Q_3 = \frac{75 \times 50}{100} = 37.5, k = 38$$

$$\therefore Q_3 = Y[38] = 1156$$

$$\begin{aligned} \text{Lower limit} &= 923 - (1.5 \times 233) \\ &= 573.5 \end{aligned}$$

$$\begin{aligned} \text{Upper limit} &= 1156 + (1.5 \times 233) \\ &= 1505.5 \end{aligned}$$

Boxplot

extreme outliers 78

mean = 1012.94

$Q_1 = 923$ $Q_2 = 1009$ $Q_3 = 1156$

